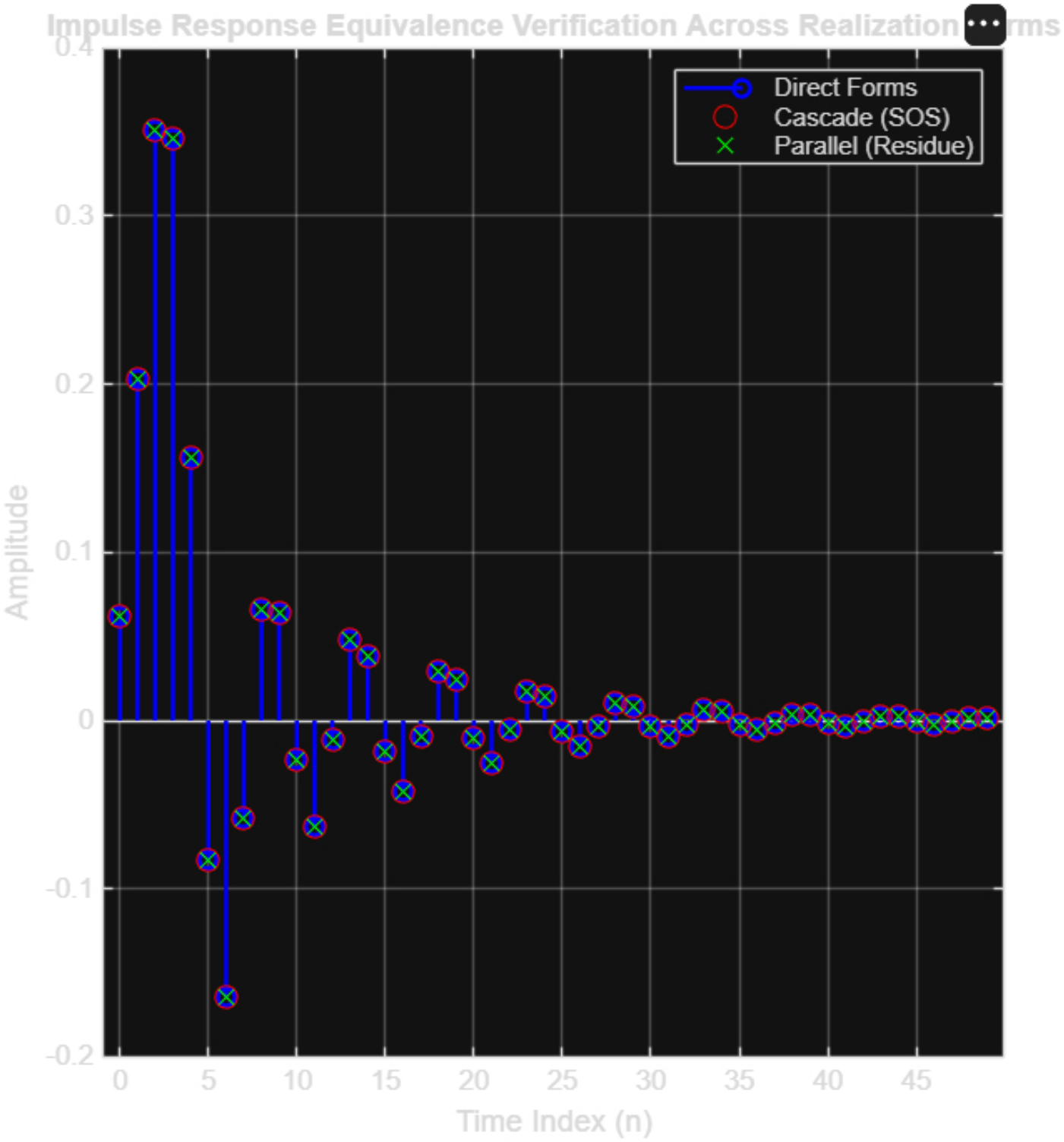
```
[b, a] = ellip(4, 1, 40, 0.4);
fprintf('============= SYSTEM COEFFICIENTS =============\n');
disp('Numerator Coefficients (b):'); disp(b);
disp('Denominator Coefficients (a):'); disp(a);
[sos, g] = tf2sos(b, a);
fprintf('\n============= CASCADE STRUCTURE (SOS MATRIX) =============\n');
disp('Each row represents [b0 b1 b2 1 a1 a2]:');
disp(sos);
fprintf('Overall Gain (g): %f\n', g);
[r, p, k] = residuez(b, a);
fprintf('\n============= PARALLEL STRUCTURE (RESIDUES & POLES) =============\n');
for idx = 1:length(r)
    fprintf('Branch %d -> Residue (r): %f + %fi, Pole (p): %f + %fi\n', ...
        idx, real(r(idx)), imag(r(idx)), real(p(idx)), imag(p(idx)));
end
fprintf('Direct Term (k): '); disp(k);
N_samples = 50;
x = [1; zeros(N_samples-1, 1)];
y_direct = filter(b, a, x);
y_cascade = filter(g, 1, x);
for i = 1:size(sos, 1)
    y_cascade = filter(sos(i, 1:3), sos(i, 4:6), y_cascade);
end
y_parallel = zeros(N_samples, 1);
if ~isempty(k)
    y_parallel = y_parallel + filter(k, 1, x);
end
for i = 1:length(r)
    y_parallel = y_parallel + filter(r(i), [1, -p(i)], x);
end
```

```
diff_cascade = max(abs(y_direct - y_cascade));
diff_parallel = max(abs(y_direct - real(y_parallel)));
fprintf('\n============ STRUCTURAL EQUIVALENCE VERIFICATION ============\n');
fprintf('Max absolute deviation (Direct vs Cascade): %e\n', diff_cascade);
fprintf('Max absolute deviation (Direct vs Parallel): %e\n', diff_parallel);
figure('Color', [1 1 1]);
stem(0:N_samples-1, y_direct, 'b', 'LineWidth', 1.5); hold on;
plot(0:N_samples-1, y_cascade, 'ro', 'MarkerSize', 8);
plot(0:N_samples-1, real(y_parallel), 'gx', 'MarkerSize', 8);
grid on;
title('Impulse Response Equivalence Verification Across Realization Forms');
xlabel('Time Index (n)');
ylabel('Amplitude');
legend('Direct Forms', 'Cascade (SOS)', 'Parallel (Residue)', 'Location', 'northeast');

```


Impulse Response Equivalence Verification Across Realization Forms
Direct Forms
Cascade (SOS)
Parallel (Residue)
Amplitude
Time Index (n)
0.4
0.3
0.2
0.1
0
-0.1
-0.2
0
5
10
15
20
25
30
35
40
45